

Bottom-Up Versus Top-Down Design

Highlights

- Bottom-up design, designing for existing work practice
- The role of biases and constraints in design
- Abstract work activities
- Top-down design, designing for an abstract work activity
- Bottom-up design and top-down design, each appropriate in different situations

12.1 INTRODUCTION

12.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 12.1). In this chapter, we look at top-down and bottom-up designs as we set the stage for the design solutions lifecycle activity.

12.2 BOTTOM-UP DESIGN: DESIGNING FOR EXISTING WORK PRACTICE

Bottom-up design is an approach to design that, avoiding insights and design ideas not derived from UX research data, begins with the details known about the work domain, work practice, and how a product or system is being used and will be used. The design is then built up in a way that will support this known usage behavior (Fig. 12.2).

The process we have covered in this book so far has been a bottom-up approach, which is predominantly a translation exercise done through a series of transformations on a path from work activity notes, to models, to UX requirements, to design.

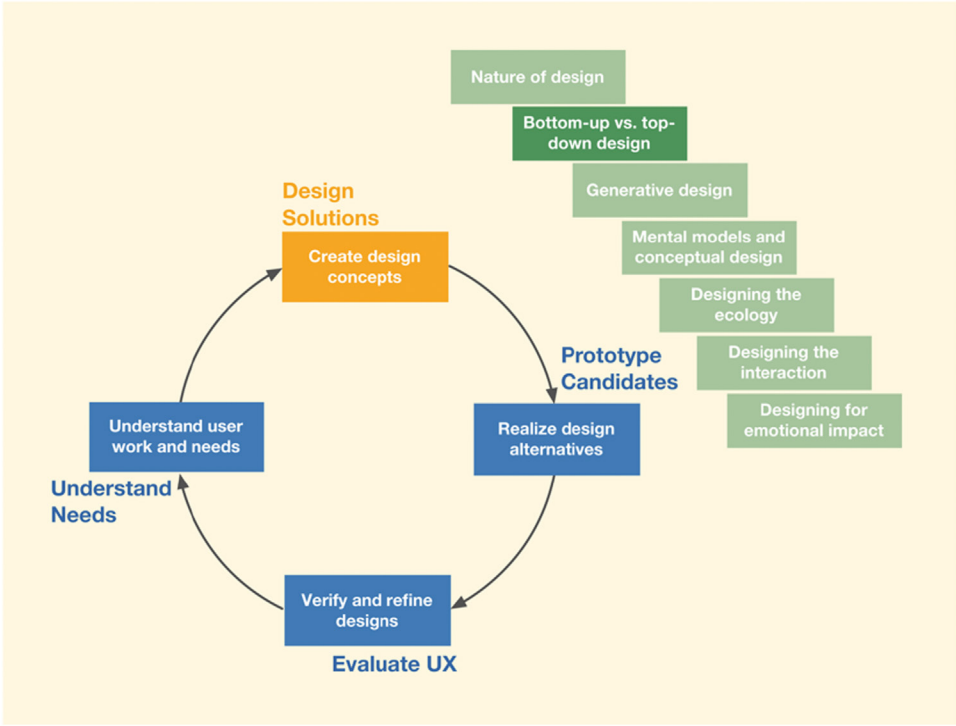


Fig. 12.1
You are here in the chapter describing two approaches to creating a UX design within the design solutions activity in the context of the overall Wheel UX lifecycle.

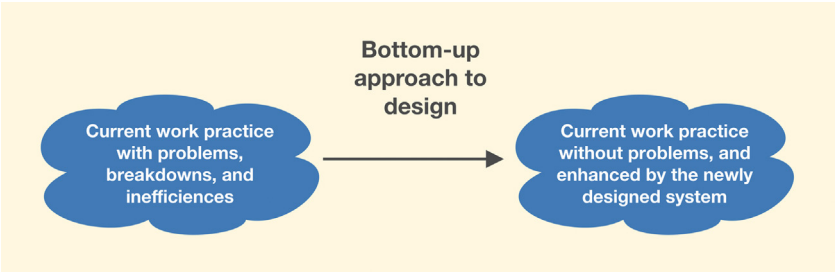


Fig. 12.2
Nature of a bottom-up approach to UX design.

12.2.1 Human-Centered Design or User-Centered Design: Common Names for Bottom-Up Design

Human-centered design (HCD) and user-centered design (UCD) are established terms in the discipline of human-computer interaction. UCD was coined by Norman and Draper (1986). To begin, HCD/UCD philosophy is based on understanding user needs and emphasizes empathy toward their goals and aspirations. It is about fitting technology to the user, rather than the user to technology.

UCD is popularly characterized as being driven by user research, what we call UX research and what others have called contextual inquiry (Beyer & Holtzblatt, 1997). It is guided by UX evaluation and iteration. In sum, UCD is a bottom-up approach based on existing work practice.

12.2.2 Designing for Existing Work Practice Is Practical

Bottom-up design, designing for existing work practice, is often the most practical approach. It does not create conflicts or friction with stakeholders because the introduction of the design solution only solves problems and removes barriers in the existing work practice. There are no disruptions in, or undue transformation of, how work currently gets done.

It is easy to understand why, when the discipline of HCI began, there was a strong push for this kind of user centeredness. There was a need to get designers (who were all engineers at that time) to prioritize users and their needs over all else. The fear was that introducing ideas not substantiated by usage data would take us to a place where designers created solutions they thought were good for users, but in fact they were not.

12.2.3 The Role of Biases and Constraints

A bias or constraint in the UX design process is an influence on how we think about UX research data, on how we collect it, and on how we view and analyze it based on something we know outside what the users tell us.

12.2.3.1 *Bias and inertia from existing usage and user preferences*

When designing for the existing work practice, the ceiling on the range of design possibilities is set through the lens of current user behavior and work practice, current revenue, current product, and current organizational biases and ways of doing things.

The bias of customer preference can subdue even an experienced designer's enthusiasm to adopt what they think is a better design. Going against such strong preference also goes against the "you are not the user" tenet and the need to have empathy toward users. Your cross-functional team, the product people, the developers, and the sales specialists all need to be convinced as to why you are going with an idea that is counter to what users say they want.

Consider the example of the Blackberry smartphone and the Apple iPhone. Before the introduction of the iPhone, the Blackberry phone was the market leader. Although the Blackberry was never known for its ease of use, it became

the smartphone of choice in the corporate world because it was the first truly capable smartphone for a working professional, plus it had excellent security and a physical keyboard.

If Apple had done UX research and analysis of existing work practice, they would have found a consistent and almost unanimous preference for a physical keyboard. This preference was so strong that it was widely predicted that Blackberry users would never adopt the iPhone, considered by many at the time to be a toy. If Apple designers had adopted a bottom-up design approach, it would have taken superhuman persuasive powers to convince the sales, engineering, and senior executives that, despite current preferences for the familiar, the touch paradigm is the future!

The designer who normally works in a framework of centering everything around satisfying customers and users will find it incredibly hard to not succumb to the biases and pressures of the usage data and other organizational and cultural trends in place. It takes a bold vision of the future to break from relying on UX research data about the existing work practice and closing your eyes to the risk.

12.2.3.2 *Bias and inertia from market success*

When a product becomes successful and popular in the market, there is a natural tendency to stick with what works and continue that same line of design with incremental refinement over time. This introduces a design complacency that can block innovation. The downfall of Blackberry is a good example of existing market success acting as a bias against continuing to be innovative in design.

Perhaps the most striking example of this kind of bias is the case of Kodak, which was the market leader in film photography and inventor of the digital camera. The work practice of film photography was less flexible and more expensive than going digital, but Kodak had bias and inertia from market success, so they did not embrace the new digital technology. Kodak executives could not bring themselves to risk their hugely successful business of selling film, paper, and other supplies for film photography, a business that had been their bread and butter for almost a century. As a result, they missed out on what could have been an amazing future. Now, Kodak is, for the most part, irrelevant in today's expansive photography market.

12.2.3.3 *Effects from advances in technology*

Existing work practice is often limited by what was possible with yesterday's technology. Advances in technology can open up work practice to a new world

of usage possibilities. The iPhone and iPod were not the first attempts at these kinds of devices (e.g., the older technology of a personal digital assistant), but they emerged when advances in technology favored increased reliability of touchscreens, reduction of device size and weight, enhanced battery life, and a great increase in storage capacity.

12.2.4 Bottom-Up Design Is Not Intended to Lead to Innovative Possibilities

While any introduction of a system into a work practice will change that practice, in bottom-up design it is only about changes to accommodate that work practice. It was never a goal of bottom-up design to see if the work practice can or should be changed at a fundamental level.

12.3 LOOKING AHEAD TO TOP-DOWN DESIGN

To create a new design that does more than just improve an existing work practice, we can approach the problem differently from the top down. To get us back in sync with where we started this discussion of work in [Section 6.2](#), we refresh your memory of some basic concepts.

12.3.1 Work

We use the term *work* to refer to the general concept of what needs to be done within a given problem domain (work domain).

12.3.2 Work Role

A work role is a user work assignment defined by the duties, functions, and work activities of a person with a certain job responsibility (e.g., customer or database administrator).

12.3.3 Work Activity

A work activity is a set of related jobs or tasks performed by users in work roles to meet a goal in the course of carrying out a work practice. Examples of work activities in a work practice include assembling an automobile, marketing a model of automobile, and using a CAD/CAM application to design an automobile.

A work activity description will have two parts:

1. The nature of what needs to be done. This is the problem, the essence of the work itself.

2. How the work gets done in a given work practice. This is the solution, a matter of procedure and protocol.

To expand on this difference, we separate these two aspects of a work activity. Because the first part is an abstraction in which the details about how the work gets done have been removed, we call this an abstract work activity.

12.3.4 Work Practice

Work practice is a complete description of how work happens in a particular context, such as an organization. It is a description of all work roles, work activities of those roles, along with the traditions, customs, and protocols associated with doing the work in that organization (e.g., a complete description of how the automobile portion of the Ford Motor company works).

12.3.5 Work Domain

An abstract, all-encompassing, and expansive description of the nature of work in a particular industry or area of business, including all the different work practices that exist in this industry, is the work domain. An example of a very large work domain is the design and production of consumer automobiles, including, for example, crash test standards and regulations.

12.3.6 Abstract Work Activity

An abstract work activity is a description of the fundamental nature of a work activity in a particular domain, stripping it to its essentials. It is the essence of that work activity involving only the central work roles, the simplest description without variations and modifications due to biases and constraints reflected in legacy designs or historical, business, political, or other influences.

As a simple but effective example, an abstract view of the act (work) of voting in a democracy is simply about people making choices among candidates. This abstract work activity can be instantiated into multiple concrete versions of the work activity, each situated within a different work practice. Each concrete work activity for voting is a solution to the abstract work activity problem of voting, as it is done in, say, one state of the United States. A concrete work activity is richer, including all work roles, and is something we try to capture in those detailed, and even overlapping, models that we build during our UX research phase (see [Chapters 8 and 9](#)). More about the voting example follows in later sections.

An example of a high-level abstract work activity can be seen in the consumer automobile domain: A manufacturer produces cars that are then sold to consumers. A corresponding high-level concrete work activity (instantiated from the abstract work activity) could include the following: Ford manufactures the car; multiple dealers stock the car; and a customer comparison-shops the car from multiple dealers, negotiates the price, buys the car, and drives it away from the lot. This work activity is part of a practice that exists that way because of historical reasons and regulations in certain US states prohibiting manufacturers from selling directly to customers.

Another concrete work activity for this same abstract work activity is a variant offered by Tesla, in which the manufacturer makes a car and stocks it, customers buy it, and the manufacturer delivers it to the customer's home (having secured legal permission to do so).

12.3.7 Abstract Work Activities Illuminate Possible New Design Targets

A bottom-up design approach will generally lead to a design target supporting existing work practice. Top-down design can lead to design targets with the potential for radically different work practice (Fig. 12.3). Using the technique of abstraction from our UX technique toolbox (see Section 2.3), we look at our descriptions of an existing work practice and strip away constraints, biases, and other contextual information that is not essential to the nature of work in that practice. We get to the core of the problem and the underlying goals and objectives of user roles in that work domain. This definition of the abstract work activities opens new paths when instantiated with new or different contextual factors, which we discuss next.

12.4 TOP-DOWN DESIGN: DESIGNING FOR ABSTRACT WORK ACTIVITIES

In a top-down approach to UX design, you begin with abstract descriptions of work activities that are stripped of information about existing work practice. Then you move toward the best design solution that enhances and supports the fundamental nature of the work, independent of current practices, preferences, traditions, or constraints.

The primary driver in the top-down approach is the designer and the designer's knowledge, skills, experience, and intuition. However, user/usage knowledge still can be, and is, used to inform design creation, but it does

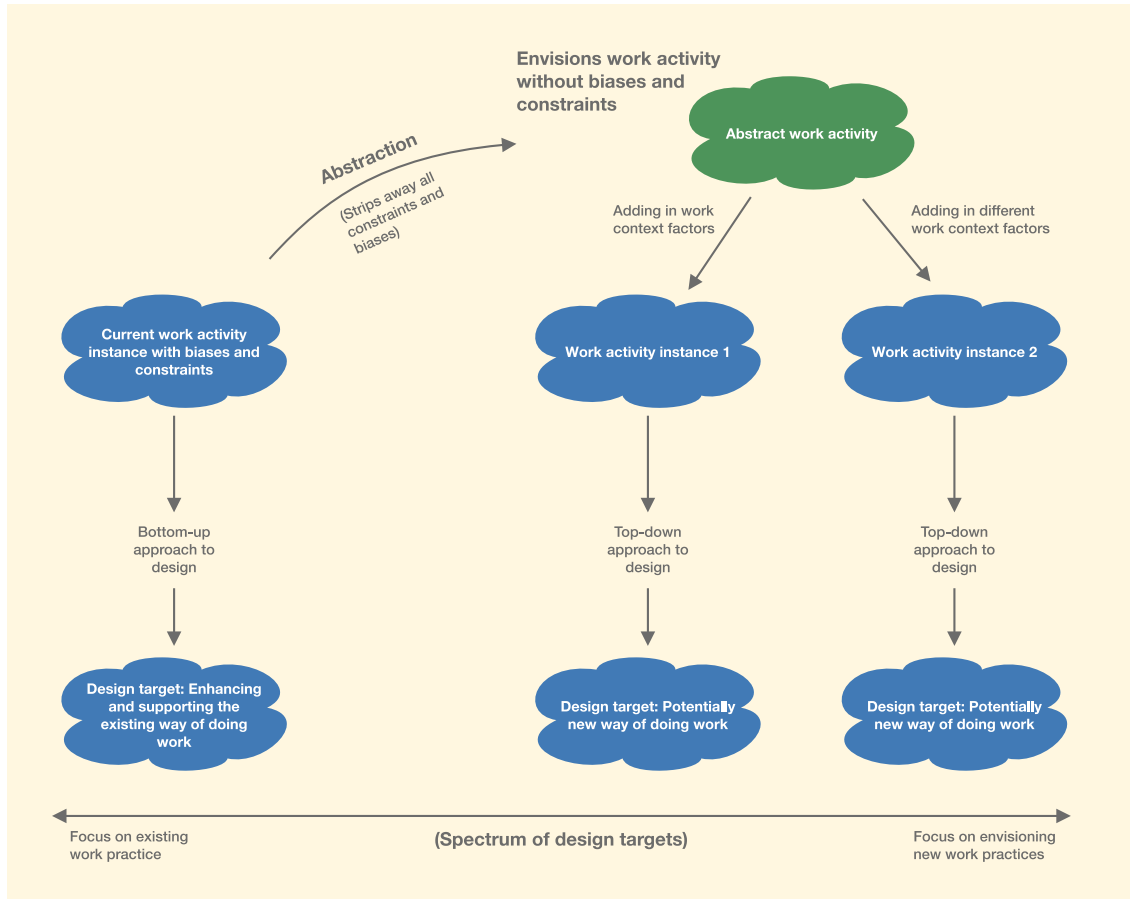


Fig. 12.3
*Different design targets:
Bottom-up design based on
existing work practice vs.
top-down design based on
an abstract work activity
description.*

not drive the process. Both approaches need UX research, but it is used in different ways:

- In bottom-up design, UX research is to analyze and model the existing work practice to improve it.
- In top-down design, UX research is to formulate and understand the abstract conceptualization of underlying essentials of the work to come up with the best overall design.

12.4.1 Consider the Example of Voting in a Democracy

To continue with our example about voting, imagine a business brief to design a voting booth for the state of Virginia in the United States. Imagine the state is in the process of updating its systems and is interested in going digital with touchscreen voting booths. This is a typical design brief, in which a client asks for a solution to support or solve specific problems facing a known practice.

Imagine a different business brief to design a voting platform, this time from the state of Alaska in the United States. Alaska uses ranked choice voting¹ of candidates, in which each voter can select and rank multiple candidates in an order of preference. This would involve additional work activities that are not present in practices like those in Virginia to tabulate and compute votes from a stack-ranked ballot.

Another work practice in the voting domain is caucusing² in US states such as Iowa, where users meet in community centers and select candidates through a show of hands (a simplified version). The work roles in such a practice would therefore be different and would match the activities involved in running a primary caucus.

12.4.1.1 The bottom-up approach

In a bottom-up approach for Virginia, we design for the existing work practice, beginning with UX research studies of voting with election booths to understand the current voting work practice in Virginia. We create work activity affinity diagrams, flow models, and other models to describe the following:

- Pre-voting preparation:
 - How citizens register to vote
 - How they find the nearest voting stations
 - What they need when they show up at the voting stations
- The onsite configuration:
 - How voting booths are set up
 - How voters interact with the booths
- The workflow of voting:
 - Voter identification
 - Checking voters off the rolls
 - Guiding voters to the appropriate booths
- Post-voting follow-up:

¹https://en.wikipedia.org/wiki/Ranked-choice_voting_in_the_United_States

²https://en.wikipedia.org/wiki/Iowa_caucuses

- Counting ballots
- Combining tallies from all sites
- Announcing winners

The design activity focuses on specifics of this particular work practice:

- Physical design of the voting booth
- Clear labels on the touchscreen, accessibility issues with the booth, color contrasts and other sensory issues, ergonomics of the booth, error avoidance and recovery in case of user mistakes during voting, materials for the booth, etc.
- Making sure it all fits with the already established ecology (the entire environment associated with voting)

In contrast, a bottom-up approach for Alaska will have to account for an entirely different workflow for voting, particularly on the need for the design to educate voters on ranked choice voting and to support them in doing it. The bottom-up approach to design for Iowa will not have much overlap with either of these two workflows because the work practice is entirely different.

12.4.1.2 *The top-down approach*

Using a top-down approach based on an abstract work activity gets us to think about the essence of this work activity: designing the best way to get eligible users to elect someone from a list of candidates. The variations of work practice across states are not relevant because the abstract work is the same.

This new framing opens up other possibilities, even beyond the existing approaches, to support this work activity, bringing into consideration all ways, including voting booths, to make this work happen. Perhaps the process can be made easier and more accessible via a smartphone app or a website to vote from the comfort of the home? Or a 1-800 number where citizens can call and cast their votes? Or an internet-connected device in the home could be used for voting and could also be used for other expressions of opinions, including petitions, surveys, and likes or dislikes of products. These new voter participation channels allow for possibilities where citizens can vote over a period of time leading up to a deadline. They even provide for flexibility, allowing citizens to change their vote after it is cast up to the deadline—something mail-in ballots or early voting practices do not allow. Perhaps new information about a candidate was unearthed in the interim.

New voter participation channels can also allow for quicker follow-up or run-off polling if required because these new ways of participation are easy

to access. Such top-down design emphasizes designing for a very different envisioned ecology (in fact, defining the new ecology is part of this approach). The resulting new work practice will need design for new interaction (workflows, etc.), information (e.g., full display of candidates and their party affiliations, which offices a given voter can cast ballots for, full description of each side of all issues and proposals up for a vote), and emotional needs after ecological design is completed.

12.4.1.3 Abstraction redirects designer thinking

Many questions can be uncovered by this new way of thinking about work. Designing for the abstract work activity can result in a system that is significantly better at supporting the act of voting. It even questions the need for a voting booth. Whereas in the bottom-up case, the discussion for the most part centered on the existing way of casting a vote.

Although it is almost certainly apocryphal, Henry Ford is reputed to have once said, “If I’d asked people what they wanted, they would have asked for a better (or faster) horse.” Regardless of whether the attribution is accurate, the quote points out the lack of regard Ford had for bottom-up inputs from customers and the faith he held in his genius at top-down design. Indeed, cars have become much more than highly improved horses and carriages. Ford knew to interpret what the users were saying not as needs and not as design solutions but as inputs to inform and inspire a new concept. The point of this example is to illustrate how thinking of the abstract work activity provides a different design target than that of a specific work activity instance.

12.4.2 Characteristics of Top-Down Design

Top-down design can be perceived as visionary. Because top-down designs are not constrained by current work practice, they can turn out startlingly different and even futuristic designs.

Top-down design is heavily driven by domain knowledge. Designers need extensive domain knowledge to abstract the nature of work in that domain.

Being a potential user in the domain is good for a designer. An important factor in the success of Apple designers practicing top-down design is the fact that they could see themselves as users of the iPhone and iPad, among its other products.

Another example of a domain in which designers can see themselves as users is photography. To design a photo editing and management application, it would help, perhaps even be essential, for designers to be experts in photography and even avid users of such a product, which would help with

immersion necessary to creatively think about the problem in an abstract manner.

12.4.3 Top-Down Design Is Not Always Practical

Sometimes, however, constraints, conventions, regulations, history, and traditions that we discussed as biases in [Section 12.2.3](#) can be too difficult to overcome in practice. They may even be decreed by law or built in to protect public safety.

Business constraints can take priority. As an example of business constraints, all parties that benefit from the current practice, such as the booth vendors, the electronic voting machine manufacturers, and perhaps even the candidates on the ballot, will resist any new ideas.

Innovation can conflict with human comfort from familiarity. Humans are not naturally open to extensive change. It unsettles comfortable routines and practices; we prefer harmony and consistency to change and disruption.

An abstract view of work activity can work against short-term goals. Because of the upheaval of the work practice, designing for the abstract work activity can be more expensive and take longer to introduce. This flies directly in the face of the unyielding push toward shorter-term returns and the reluctance to tackle larger projects, a mindset of business these days.

Technology constraints can limit innovation. Sometimes the requisite technology is not quite ready. Ascertaining voter eligibility, preventing fraud, and ensuring voters' choices are not personally identifiable after votes are cast are all difficult technical problems. UX designers must do a thorough exploration of the available technology (Gajendar, 2012).

12.4.4 Extreme Top-Down Design Is the Path to Disruptive Design

On the positive side, top-down design can be disruptively innovative. Sometimes an innovative design comes along at the right time and place in such a way that the benefits of getting past the upheaval outweigh the discomfort of going through change. In such cases, top-down design can result in an innovative next-generation product (e.g., the iPhone). Another way to put it is, a good bottom-up design fills a niche, but a good top-down design can create a niche (Moore, 2017).

Example: Top-Down Design in Architecture

Frank Lloyd Wright is an archetypal example of a disruptive top-down designer in architecture. He was supremely confident that he, as designer,

knew best what was good for the user. Wright and Costner agree, “If you build it, they will come.” Wright took top-down design to the extreme. It was no longer about what users thought: “I don’t build a house without predicting the end of the present social order. Every building is a missionary... It’s their [the users’] duty to understand, to appreciate, and conform insofar as possible to the idea of the house” (Lubow, 2009). This is about as far from UCD as you can imagine.

12.5 SEGUE

In this chapter, we contrasted the two approaches to design to either support existing work practice or to fundamentally transform it. In the next chapter, we dive into the business of generating design ideas, where we will talk about ideation, sketching, and critiquing.